Foundations of GenAI Laboratory

ShopAI: AI-Based Product Description and Attribute Generator for Shopify Merchants Using BLIP and Computer Vision

SDG's:SDG 8 – Decent Work and Economic Growth,
SDG 9 – Industry, Innovation and Infrastructure ,
SDG 12 – Responsible Consumption and Production

Abstract: ShopAI converts one product photo into an editable Shopify listing: a title, five visual attributes, a description and key features. A local FastAPI service runs the pretrained Salesforce BLIP captioning model; its caption is combined with Pillow-based colour analysis and keyword rules to fill each field. The system never invents material, size or performance claims, and marks uncertain attributes with a confidence level so the merchant can verify them. A vanilla HTML/CSS/JavaScript front end offers sign-in, an IndexedDB product catalogue and an edit view. Analysing one image takes about 4 to 10 seconds on CPU, and the 10 extractor unit tests pass.

Objectives:
1. Generate a title, description and key features from a single product image.
2. Extract category, colour, shape, pattern and texture with confidence labels.
3. Avoid fabricating materials, sizes or performance claims.
4. Let merchants review, edit and save listings in a simple web app.

Dataset
Type of dataset: none for training; pretrained image-captioning model plus user-uploaded product images (JPG/PNG/WEBP, up to 10 MB).
Source: https://huggingface.co/Salesforce/blip-image-captioning-base
Attributes: category, main colour, shape/style, pattern, visible texture (generated outputs).

Software (Front, Back): Front: HTML5, CSS3, vanilla JavaScript, IndexedDB. Back: Python 3.10, FastAPI, Uvicorn, PyTorch.
Library: transformers, torch, pillow, pydantic, numpy, python-multipart, pytest.
Functions Used: load_model(), generate_caption(), validate_and_open_image(), extract_dominant_color(), reconcile_color(), extract_category/shape_style/pattern/visible_texture(), build_product_analysis(), generate_description() API route.

Deep Learning Algorithm: BLIP vision-language model (ViT image encoder with transformer text decoder, beam search with 3 beams) for captioning, followed by deterministic colour segmentation and rule-based attribute extraction.

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